

Instructor: Date:

Learner: Regional Branch:

Program: Center:

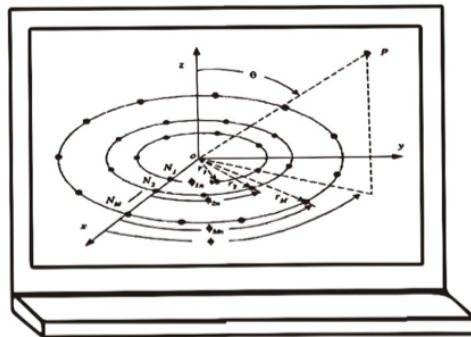
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Let's Start!

Are you drawing a disorganized diagram?



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can talk about the use of sequence diagrams.
- I can talk about the objects that appear in a sequence diagram.
- I can identify concepts like selection and iteration in sequence diagrams.

Ask your partner:

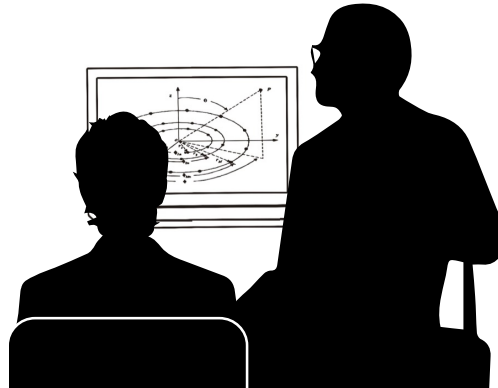
- Why do you think diagrams have to be dynamic?
- Do you think sequence is important when drawing a diagram? Why?





Let's Explore!

Manager is proposing a particular scenario to understand a withdrawing process



Instructions: Read the text below.

Employee: Are you representing this process in a diagram?

Manager: Absolutely! I'm representing it by using sequence diagrams.

Employee: Is it describing the architecture or the state of a system?

Manager: No, sequence diagrams don't describe that. We're using them to describe how a system works over a period of time. They aren't static, but dynamic.

Employee: Are they showing the sequence of method invocations?

Manager: Exactly! But between objects over a period of time. They let us know how objects collaborate in the particular scenario I'm proposing.

Employee: What is the diagram showing?

Manager: We have two objects within a scenario where someone wants to withdraw money from an ATM. Vertical dashed lines show that time runs from top to bottom.

Employee: Are still objects indicating actions for a long time after the loop?

Manager: Sure! They continue expressing actions that take place for as long as necessary.

Employee: So, the user inserts his card into object1-ATM, during the same period of time Object1-ATM needs to verify it with the bank to find the user's account. The bank finds it and returns a value to let the process continue. Object1-ATM receives a message and it shows it to the user, then, ATM allows him to withdraw his money.

Employee: What other aspects are you taking into account?

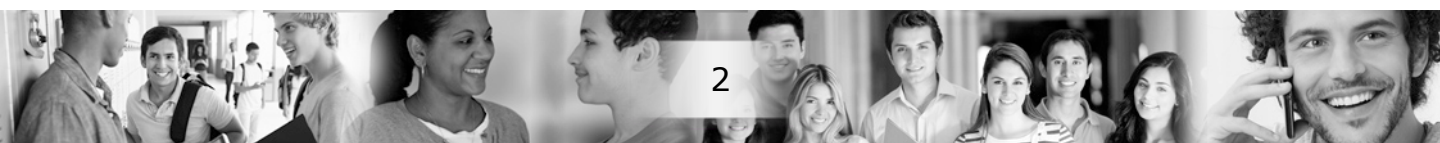
Manager: Remember 'ifs' and 'loops'

Employee: Do they give logic to the scenario?

Manager: Exactly! The logic of a scenario depends on them, the selection by using 'ifs' and iteration 'loops'.

Employee: Are they making the diagram simpler?

Manager: No. They clutter sequence diagrams, because we need interaction frames for representing them in sequence diagrams.



**Let's Practice!****Selection and Iteration in Sequence Diagrams**

Instructions: Circle the correct answer. Read the sentences aloud. Check your pronunciation with your instructor.

1. Sequence diagrams are
 - a. static
 - b. dynamic
 - c. neither
2. Selection is represented by it.
 - a. if
 - b. loops
 - c. dashed lines
3. Loops help you to model a repetitive sequence, it represents:
 - a. selection
 - b. interaction
 - c. methods

**Let's Practice!**

Let's review our concepts on sequence diagrams.

Instructions: Underline the correct option to correctly complete each sentence. Use the questions to model a conversation with the help of your instructor. Take turns to read the questions with your partner.

1. Are you (representing / represents / represent) the process of creation in a diagram?
2. Is the sequence diagram (describing / described / describe) the architecture or the state of a system?
3. Are these sequence diagrams (showing / show / shows) the sequence of method invocations?
4. Are sequence diagrams (giving / gived / gives) logic to the scenario?
5. Are *loops* and *ifs* (making / make / maked) the sequence diagram easier to understand?





Extension Activity

Interpretation of a sequence diagram.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search examples of sequence diagrams.
- Choose one and interpret it.
- Write your interpretation in a piece of paper.
- Share with your class

Self-Assessment

Yes No

- I identify sequence diagrams.
- I interpret sequence diagrams.
- I explain my interpretation of the sequence diagram.

☐	☐
☐	☐
☐	☐



Target Words

- **loop** (noun): circuit
e.g. The logic of a scenario depends on them, selection 'ifs' and iteration 'loops'.
- **interaction frame** (noun): unit of behavior that represents a notion or an interaction
e.g. They clutter sequence diagrams, because we need interaction frames for representing them in sequence diagrams.
- **clutter** (verb): to disorganize
e.g. They clutter sequence diagrams, because we need interaction frames for representing them in sequence diagrams.
- **method invocation** (noun) : objects that can interact on different diagrams but with the same network
e.g. Are they showing the sequence of method invocations?
- **iteration** (noun) : repetition
e.g. The logic of a scenario depends on them, selection 'ifs' and iteration 'loops'.
- **dashed line** (noun) : dashes you use to make a line
e.g. Vertical dashed lines show that time runs from top to bottom.



Learning Points

Present Continuous in Questions

You use **questions in present continuous** to ask for information that is set in a specific moment in present.

For making **interrogative** statements you need to follow this grammatical structure:

Am / Is / Are + Subject + Verb-ing + Complement + ?

Example:

Are you using a UML diagram?

Besides yes/no questions, you have also the possibility of asking information questions. These kind of questions use a Wh- word in order to center on a specific piece of information, such as **what** (a specific object or piece of information); **when** (time); **how** (manner); **where** (place); **why** (reason); and **which** (choice).

The structure for this kind of questions is very simple:

Wh- + Verb to be + Subject + Verb - ing *+ Complement + ?

Example:

What are you planning to do with that information?

How are you planning to organize it?

Examples

- *Is it describing the architecture or the state of a system?*
- *How are you representing this process in a diagram?*



Learning Points

Subject and object pronouns

You use **subject and object pronouns** to replace subject or objects within a statement.

You use subject pronouns when we want to reduce proper or common nouns repetition within statements.

Example:

The programmer is drawing the new sequence diagram. **He** (he = the programmer) is drawing the new sequence diagram.

You use object pronouns when you want to replace the object or who receives the action.

Example:

He is drawing the new **sequence diagram**.

He is drawing **it** (the new sequence diagram).

You can always use object pronouns after a preposition e.g. **for him**

Subject Pronoun	Object Pronoun
I	Me
You	You
He	Him
She	Her
It	It
We	Us
They	Them

Examples

- They clutter sequence diagrams, because we need interaction frames for representing them in sequence diagrams.
- ATM allows him to withdraw his money.
- We're using them to describe how a system works over a period of time. They aren't static, but dynamic.